

Bandon Public Transport Accessibility Analysis

Bandon, County Cork, Republic of Ireland

SEL-002 | Public Transport Accessibility & Vulnerability

Prepared as a reproducible GIS-based spatial evidence study

Contents

Contents.....	2
Abstract.....	4
Research Questions.....	4
Executive Summary.....	4
Answers to the Research Questions.....	5
1. Understanding Public Transport Accessibility.....	6
1.1 Public Transport Accessibility in Urban and Regional Planning.....	6
1.2 Accessibility as a Spatial Concept.....	6
1.3 Accessibility in Small Towns.....	6
1.4 Accessibility and Social Equity.....	7
1.5 Research Context.....	7
2. Study Area and Analytical Framework.....	8
2.1 Study Area Overview.....	8
2.2 Study Context and Population Vulnerability Assessment.....	8
2.3 Research Questions and Objectives.....	9
2.4 Analytical Framework and Map Series.....	9
3. Dimensions and Methodology.....	10
3.1 Walking Accessibility (Map 01.01).....	10
3.2 Distance-Based Accessibility (Map 01.02).....	10
3.3 Service Availability (Map 01.03).....	10
3.4 Population Vulnerability (Map 01.04).....	10
3.5 Composite Accessibility Index (Map 01.05).....	10
3.6 Equity-Focused Vulnerability Hotspot Model (Map 01.06).....	10
3.7 Data Infrastructure and Sources.....	10
3.8 Methodology and Workflow.....	11
3.9 Walking Accessibility Analysis.....	11
3.10 Service Frequency Analysis.....	11
3.11 Vulnerability Index Construction.....	11
3.12 Composite Accessibility Index.....	12
3.13 Hotspot Analysis and Equity-Led Prioritisation Framework.....	12
4. Results and Spatial Interpretation.....	13
4.1 Map 01.00 - Accessibility Study Area.....	13
4.2 Map 01.01 - Walking Catchments.....	14

4.3 Map 01.02 - Walking Distance Score.....	15
4.4 Map 01.03 - Peak-Time Service.....	16
4.5 Map 01.04 - Population Vulnerability.....	17
4.6 Map 01.05 - Composite Accessibility Index (CAI).....	18
4.7 Map 01.06 - Equity-focused vulnerability hotspots.....	19
4.8 Cross-Map Synthesis.....	19
5. Evaluation of Research Questions.....	20
5.1 Research Question One - Viability for Cork City Commuters.....	20
5.2 Research Question Two - Identifying Vulnerable Areas and Groups.....	21
5.3 Spatial Typology of Bandon.....	21
6. Policy and Planning Implications.....	22
6.1 Priority Intervention Areas.....	22
6.2 Pedestrian Infrastructure Improvements.....	22
6.3 Service Frequency Optimisation.....	22
6.4 Bus Stop Expansion and Network Coverage.....	22
6.5 Accessibility Monitoring.....	22
6.6 Recommendations for Cork County Council and NTA.....	22
7. Limitations and Future Research.....	23
7.1 Limitations.....	23
7.2 Future Research Priorities.....	23
8. Conclusion.....	24
8.1 Summary of Findings.....	24
8.2 Research Contribution.....	24
8.3 Final Reflection.....	24
References.....	25
Appendix A. Data Sources and Reproducibility Register.....	25
Appendix B. Map Production Register.....	25
Document Control.....	25

Abstract

Bandon is one of the fastest-growing towns in West Cork and occupies a distinctive position within the regional settlement hierarchy. According to Census 2022, the town has a resident population of 8,196 and functions simultaneously as a commuter town within the Greater Cork metropolitan area and as a gateway to the wider West Cork region. Its strategic location approximately 30 km southwest of Cork City makes it an important origin and destination for daily commuting, education, healthcare, retail, and regional services.

This dual role raises critical questions about the performance of the public transport network. While Bandon is connected to Cork City and surrounding settlements through regional bus services, the extent to which those services provide an effective and equitable alternative to private-car travel varies across the town. Accessibility is therefore considered not only in terms of the physical location of bus stops, but also in relation to the characteristics and needs of the population living within different parts of Bandon.

Research Questions

Is public transport a viable alternative to private-car commuting between Bandon and Cork City?

Which areas are potentially most vulnerable to limited public transport accessibility because of their demographic, health, employment, or household characteristics?

The first question focuses on the spatial performance of the public transport system through walking accessibility and service frequency. The second recognises that accessibility should be evaluated within a broader social context, acknowledging that some population groups are more likely to depend on public transport than others. A bus stop may be geographically accessible, yet still fail to provide meaningful accessibility if services are infrequent or if populations with greater mobility needs remain poorly served.

Executive Summary

The hotspot analysis demonstrates that public transport accessibility in Bandon is spatially uneven and strongly centralised. Areas surrounding the town centre benefit from the highest levels of accessibility due to their proximity to the four principal bus stops and the concentration of scheduled bus services. In contrast, several peripheral residential neighbourhoods experience substantially lower accessibility while simultaneously containing populations with characteristics associated with greater transport vulnerability.

The findings indicate that accessibility and vulnerability do not follow the same spatial pattern. Some neighbourhoods have relatively low accessibility despite comparatively low vulnerability, while others combine limited accessibility with demographic and socio-economic characteristics that increase potential dependence on public transport. These latter locations emerge as Critical and High Priority Accessibility-Vulnerability Hotspots.

The analysis provides a spatial evidence base for identifying areas where improvements to walking infrastructure, bus stop provision, service frequency, or broader transport planning interventions could deliver the greatest social benefit.

Answers to the Research Questions

Is public transport a viable alternative to private-car commuting between Bandon and Cork City?

Answer: Partially yes. Public transport is a viable commuting alternative for residents living within the central accessibility corridor where walking access and service availability are strongest. Viability declines across several peripheral residential neighbourhoods where accessibility is lower and walking access is substantially longer.

Which areas of Bandon are potentially most vulnerable to limited public transport accessibility?

Answer: Critical and High Priority Small Areas are located primarily around the outer residential neighbourhoods, where lower accessibility combines with older age profiles, disability or poor health, retirement or unemployment, and households without private-car access.

1. Understanding Public Transport Accessibility

Public transport accessibility describes the ease with which people can reach and use public transport and, through it, access destinations and opportunities. In this study, accessibility is treated as a spatial and multidimensional concept rather than a single distance measure.

1.1 Public Transport Accessibility in Urban and Regional Planning

Accessibility is concerned with the relationship between people, transport systems and opportunities. A location may be physically close to a bus stop but still have relatively weak effective accessibility if services are infrequent. Conversely, a location with a somewhat longer walk may have stronger effective accessibility where frequent services are available. The Bandon analysis therefore separates walking access from service availability before combining them in the composite analysis.

- **Opportunities:** Accessibility links transport provision to access to employment, education, health-care, services and other opportunities.
- **Mobility vs. Accessibility:** Mobility concerns movement itself, whereas accessibility concerns the opportunities that movement enables.
- **Quality of Life:** Accessibility is relevant to quality of life because transport conditions influence people's ability to participate in social and economic life.
- **Equity:** Transport equity requires consideration of both the distribution of service provision and the distribution of population need.

1.2 Accessibility as a Spatial Concept

The Bandon framework calculates accessibility through several related spatial dimensions: distance, time, opportunity, network connectivity and service availability. Walking distance and network travel time describe the physical effort required to reach a stop, while GTFS-derived service frequency introduces a temporal dimension.

- **Distance:** How far a residential area is from a public transport stop through the pedestrian network.
- **Time:** The estimated walking time required to reach the transport network.
- **Opportunity:** The potential to use public transport once a stop is reached.
- **Network Connectivity:** The pedestrian routes available for reaching stops rather than relying solely on straight-line distance.
- **Service Availability:** The frequency and temporal intensity of public transport services derived from GTFS feeds.

1.3 Accessibility in Small Towns

Smaller commuter towns can display a sharp transition between a relatively well-served urban core and lower-density peripheral residential areas. Bandon provides a useful case study because the mapped results show that accessibility is spatially structured rather than uniform across the study boundary.

1.4 Accessibility and Social Equity

Transport disadvantage becomes particularly important where weaker accessibility overlaps with population characteristics associated with greater potential dependence on public transport. The vulnerability layer is therefore not treated as a measure of transport accessibility itself. It is a separate population-need dimension that is combined with accessibility at the hotspot stage.

1.5 Research Context

The Bandon study forms part of the Spatial Evidence Lab portfolio and applies a reproducible GIS approach to a small-town transport question.

Accessibility → Mobility → Opportunity → Quality of Life

2. Study Area and Analytical Framework

The purpose of the study is to assess how public transport accessibility varies spatially across Bandon and to identify locations where lower accessibility coincides with greater population vulnerability.

2.1 Study Area Overview

The study area is defined by the mapped study boundary around Bandon and its surrounding residential and settlement areas. The context map identifies public transport stops and the local road network. The mapped extent captures the town centre and peripheral areas including Gully, Cloughmacsimon, Kilbrogan, Ballylangley, Currough, and Knocknagrohan.

The study uses Small Areas as the principal reporting geography for population vulnerability and spatial comparison. Public transport stops are represented as point features, while the road and pedestrian network provides the spatial context for network-based access analysis.

2.2 Study Context and Population Vulnerability Assessment

Bandon has a resident population of 8,196 (Census 2022) and functions as a key regional settlement and commuter node. Bandon was selected because its defined urban core and surrounding lower-density residential peripheries make it an ideal small-town case study for examining the transition between high and lower public transport accessibility.

Population Vulnerability as a Complementary Dimension

To evaluate transport equity, the analysis incorporates Census 2022 Small Area data to construct a Population Vulnerability Index. This index does not measure accessibility itself; instead, it identifies characteristics that may increase potential dependence on public transport or indicate greater vulnerability to transport disadvantage.

Census Variable	Description	Relevance to Vulnerability Assessment
T1_1AGE70_74T	Population aged 70-74	Identifies older residents who may have increased dependence on accessible transport.
T1_1AGE75_79T	Population aged 75-79	Represents populations with potentially greater mobility needs.
T1_1AGE80_84T	Population aged 80-84	Captures areas with higher concentrations of elderly residents.
T1_1AGEGE_85T	Population aged 85+	Represents population most likely to require highly accessible services.
T1_1AGETT	Total population	Denominator for population-based vulnerability measures.
T4_5RP / T8_1_RT	Retired persons	Represents populations less likely to rely on work travel but dependent on local transport.
T8_1_LTUT	Long-term unemployed	Represents socio-economic vulnerability and transport disadvantage.
T8_1_UTWSDT	Permanent sickness/disability	Represents populations experiencing greater mobility constraints.
T12_1_T	Persons with a disability	Measures populations requiring accessible transport infrastructure.
T12_3_BT	Bad general health	Indicates communities with health-related transport needs.
T12_3_VBT	Very bad general health	Identifies areas with high health-related transport vulnerability.
T15_1_NC	Households without a car	Households potentially most dependent on public transport.
T11_1_BUT	Travelling by bus/minibus	Provides evidence of existing public transport use.
T11_2_T1-T8	Departure time for work/school	Captures commuting demand during morning departure periods.
T11_2_NS / T11_2_T	Leaving time not stated / Total	Denominator for analysing commuting time distributions.

Dimensions of Population Vulnerability

These Census variables are grouped into four core vulnerability domains:

1. Demographic Vulnerability: Older population (70+ years), retired persons.
2. Health and Mobility Vulnerability: Disability, permanent sickness, bad and very bad general health.
3. Socio-Economic Vulnerability: Long-term unemployment and retirement-related indicators.
4. Transport Dependence: Households without a motor car, bus commuters, and morning departure-time patterns.

Integration Framework

The analysis strictly separates the spatial performance of the transport network from population demand and vulnerability:

Census Variable	Purpose in the Study
Public Transport Supply & Accessibility	Measured via bus stop locations, walking-network catchments, and peak service frequency
Observed Travel Behaviour	Represented by Census commuting patterns and bus-use variable
Population Vulnerability	Represented by age, retirement, unemployment, disability, and health indicators
Private Vehicle Availability	Represented by households without access to a motor car

2.3 Research Questions and Objectives

Primary Research Question: **How can GIS-based accessibility and vulnerability analysis identify areas of transport disadvantage in Bandon?**

Research Objectives:

1. Map the spatial coverage of public transport stops using walking-network analysis.
2. Measure walking-distance accessibility to public transport stops.
3. Assess peak-time service availability using GTFS-derived service frequency.
4. Construct a population vulnerability layer using selected Census 2022 indicators.
5. Combine accessibility and vulnerability to identify spatial hotspots and priority areas.

2.4 Analytical Framework and Map Series

Map	Title	Subtitle
01.00	Bandon Public Transport Accessibility Study Area	Study boundary, public transport stops and local road network
01.01	Bandon Public Transport Accessibility - Walking Catchments	Areas within 5, 10 and 15 minutes walking distance of public transport stops
01.02	Bandon Public Transport Accessibility - Walking Distance Score	Spatial accessibility to public transport stops based on walking distance
01.03	Bandon Public Transport Accessibility - Peak-Time Service	Frequency-adjusted accessibility to public transport during peak travel periods
01.04	Bandon Population Vulnerability	Spatial distribution of population vulnerability based on Census 2022
01.05	Bandon Composite Public Transport Accessibility Index	Integrated assessment of walking access and public transport service frequency
01.06	Bandon Public Transport Accessibility - Equity-focused vulnerability hotspots	Areas where public transport accessibility and population vulnerability indicate differing levels of spatial priority

3. Dimensions and Methodology

The analytical pipeline processes accessibility through progressive dimensions:

Walking Access → Distance Score → Service Frequency → Vulnerability → Integrated Priority

3.1 Walking Accessibility (Map 01.01)

Evaluates 5, 10, and 15 minute network catchments around stops. Network routing provides a true representation of pedestrian access over straight-line buffers.

3.2 Distance-Based Accessibility (Map 01.02)

Classifies continuous distance scores into Low, Moderate, High, and Very High accessibility.

3.3 Service Availability (Map 01.03)

Adds scheduled peak-time service frequency to differentiate physical proximity from actual service utility (Low, Medium, High).

3.4 Population Vulnerability (Map 01.04)

Mapped into five relative classes (Very Low, Low, Moderate, High, Very High) across Small Areas.

3.5 Composite Accessibility Index (Map 01.05)

Synthesises walking access and frequency into the Composite Accessibility Index (CAI), classified as Very Poor, Poor, Moderate, Good, and Excellent.

3.6 Equity-Focused Vulnerability Hotspot Model (Map 01.06)

Overlays CAI and Vulnerability to create the decision-support prioritisation matrix:

Priority Category	Spatial & Social Interpretation
Critical Hotspot	Very low accessibility combined with high population vulnerability
High Priority	Low accessibility with elevated population vulnerability
Moderate	Intermediate accessibility and vulnerability scores
Low Priority	Lower vulnerability despite reduced physical accessibility
Well Served	High transport accessibility relative to population vulnerability

3.7 Data Infrastructure and Sources

The study integrates NTA GTFS schedule feeds, CSO Census 2022 Small Area statistics, OpenStreetMap pedestrian networks, and Tailte Éireann boundary data:

Data Source	Contribution to the Framework
GTFS Stops, Routes & Trips (NTA)	Public transport locations, routing, and peak timetable
OpenStreetMap	Pedestrian network routing and street topology
CSO Census 2022	Socio-demographic, health, age, and car-ownership indicators
Small Areas - Tailte Éireann / CSO	Primary spatial reporting geography

3.8 Methodology and Workflow

1. Network Processing: Build a connected pedestrian routing network in GIS.
2. Walking Catchment Generation: Calculate 5, 10, and 15 minute service areas from GTFS stops.
3. Temporal Frequency Calculation: Extract peak morning trips from GTFS stop times and calculate service frequency scores.
4. Vulnerability Normalisation: Standardise Census indicators (Elderly_Norm, Health_Norm, NoCar_Norm, Econ_Norm) into Vulnerability_Index.
5. Composite Indexing: Combine walking access scores and service frequency into Composite Accessibility Index (CAI).
6. Equity-Focused Vulnerability Hotspot Model : Matrix overlay of CAI and Vulnerability Index to generate Map 01.06.

3.9 Walking Accessibility Analysis

Walking accessibility is modelled using the pedestrian network rather than straight-line distance. Network service areas are generated around public transport stops and expressed using 5, 10 and 15 minute walking thresholds. A distance-based score is then derived to provide a more continuous accessibility representation.

The current map outputs establish the thresholds and resulting classes but do not document the precise walking-speed parameter. That parameter should be retained in the processing metadata and methodology record.

3.10 Service Frequency Analysis

GTFS schedule data are used to represent the temporal dimension of public transport accessibility. Trips and stop-time information are processed to estimate peak-time service frequency, which is converted into a frequency score and used in the frequency-adjusted accessibility map.

The exact peak-time windows and frequency calculation formula are not visible in the supplied map outputs and should be retained in the technical methodology documentation.

3.11 Vulnerability Index Construction

A Population Vulnerability Index is constructed from selected Census 2022 indicators. The indicators are standardised and combined to create a spatial measure of relative vulnerability. The resulting geography is classified into five levels from Very Low to Very High.

The supplied report does not list the individual Census variables, their weights or their exact standardisation formula. These should be added to the project's metadata/indicator dictionary rather than reconstructed from the map.

3.12 Composite Accessibility Index

The Composite Accessibility Index integrates the principal accessibility dimensions represented by walking access and service frequency. The resulting CAI is classified into five levels: Very Poor, Poor, Moderate, Good and Excellent.

The CAI should be interpreted as a relative spatial index. The precise weighting and mathematical formula should be documented in the reproducibility appendix and linked to the processing script or SQL transformation that generated the final field.

3.13 Hotspot Analysis and Equity-Led Prioritisation Framework

The Equity-Focused Vulnerability Hotspot Model (Map 01.06) identifies priority areas by integrating physical transport accessibility deficits with socio-demographic vulnerability. Rather than evaluating physical access in isolation, this approach adopts an equity-led prioritisation framework to ensure that public transport and pedestrian infrastructure investments target communities with the greatest potential transport dependency.

- **1. Core Model Variables**

The hotspot formulation relies on two continuous, normalized spatial variables defined at the Small Area level:

Access Deficit: Quantifies the spatial deficit in transit access, calculated as:

$$\text{Access Deficit} = 1 - \text{Normalised CAI}$$

Higher values denote poorer transit accessibility and longer network walking distances.

VUL norm: Represents the composite population vulnerability score derived from standardized Census 2022 indicators. Higher values reflect higher concentrations of sensitive demographic groups (e.g., elderly residents, car-less households, long-term unemployed, and persons with mobility constraints or poor health).

- **2. Mathematical Formulation & Weighting Schema**

To operationalise social equity, the primary prioritisation model assigns a 60% weighting to population vulnerability and a 40% weighting to physical accessibility deficit:

$$\text{Hotspot Score Equity} = 0.4 \times \text{Access Deficit} + 0.6 \times \text{VUL norm}$$

- **3. Policy Rationale and Spatial Interpretation**

By placing greater weight on population vulnerability (60%), the model prevents high-priority ranks from being assigned solely to unpopulated or low-density peripheral areas that happen to have long physical walking distances. Instead, higher vulnerability exerts a stronger pull on the final score. A location becomes a Critical or High Priority Hotspot specifically where limited physical accessibility intersects with populations exhibiting acute transit reliance.

- **4. Benchmark Comparison: Balanced Model**

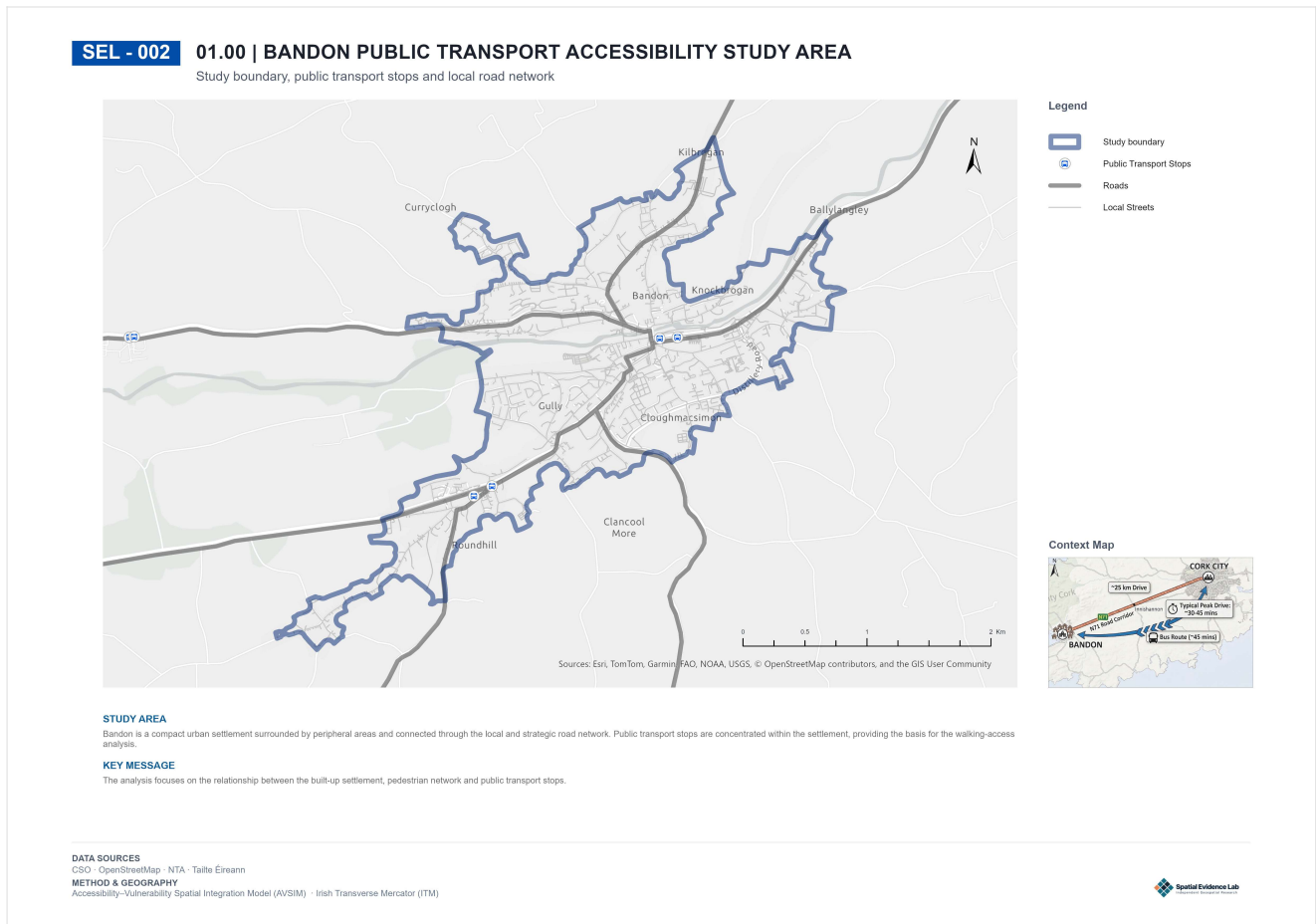
For sensitivity analysis and policy evaluation, the dataset includes a secondary, balanced benchmark model (50/50 split):

$$\text{Hotspot Score Balanced} = 0.5 \times \text{Access Deficit} + 0.5 \times \text{VUL norm}$$

4. Results and Spatial Interpretation

The map series provides an immediate visual reading of the spatial structure of public transport accessibility in Bandon. The most consistent pattern is a stronger accessibility core around the town centre and weaker conditions toward peripheral areas.

4.1 Map 01.00 - Accessibility Study Area



Description

This map establishes the spatial context for the accessibility assessment, showing the Bandon study boundary, public transport stops and the surrounding road network. It identifies the relationship between the built-up settlement, the location of public transport infrastructure and the peripheral residential areas that are subsequently assessed through network-based accessibility analysis.

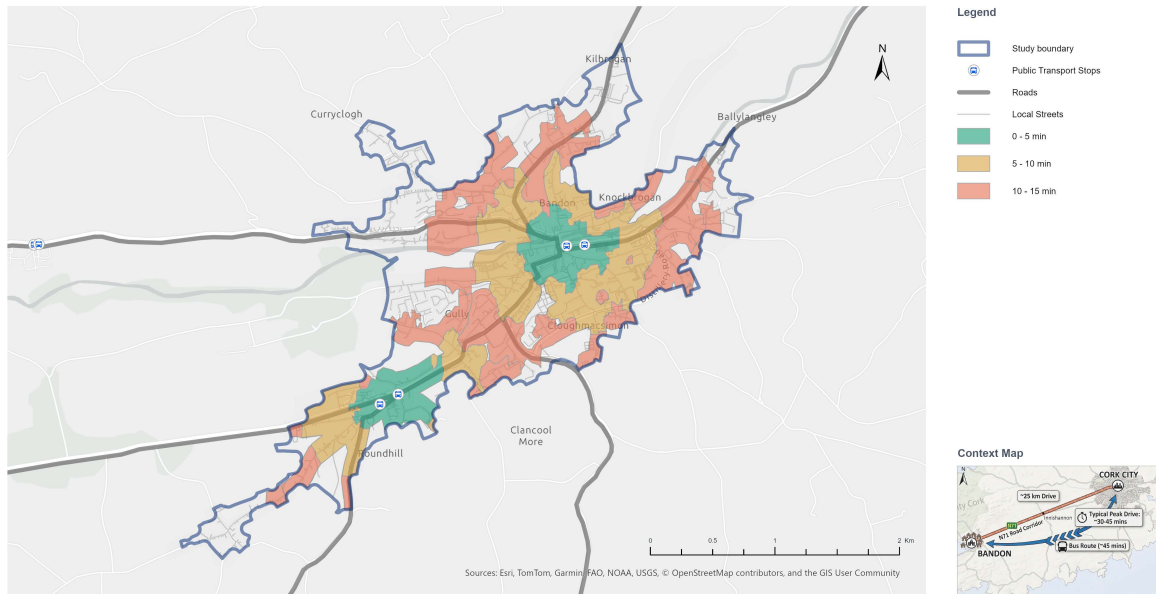
Key Message

Bandon's public transport network is concentrated within the settlement, providing the spatial foundation for assessing how accessibility changes from the town centre towards peripheral residential areas.

4.2 Map 01.01 - Walking Catchments

SEL - 002 01.01 | BANDON PUBLIC TRANSPORT ACCESSIBILITY - WALKING CATCHMENTS

Service areas within 5, 10 and 15 minutes walking distance of public transport stops



STUDY AREA

Bandon is a compact urban settlement with public transport services concentrated within the built-up area, while peripheral areas extend beyond the main service network. The analysis examines how pedestrian access to public transport varies across the settlement and its surrounding areas.

KEY MESSAGE

Walking distance reveals a clear core-to-periphery accessibility gradient, with Very High and High accessibility concentrated around well-served areas and Low accessibility increasing toward the settlement's periphery.

DATA SOURCES

CSO - OpenStreetMap - NTA - Taithe Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map measures the effective pedestrian reach of public transport using network-based 5-, 10- and 15-minute walking catchments around public transport stops. Unlike straight-line buffers, the catchments reflect the available pedestrian network and therefore provide a more realistic representation of how far residents can walk to access public transport.

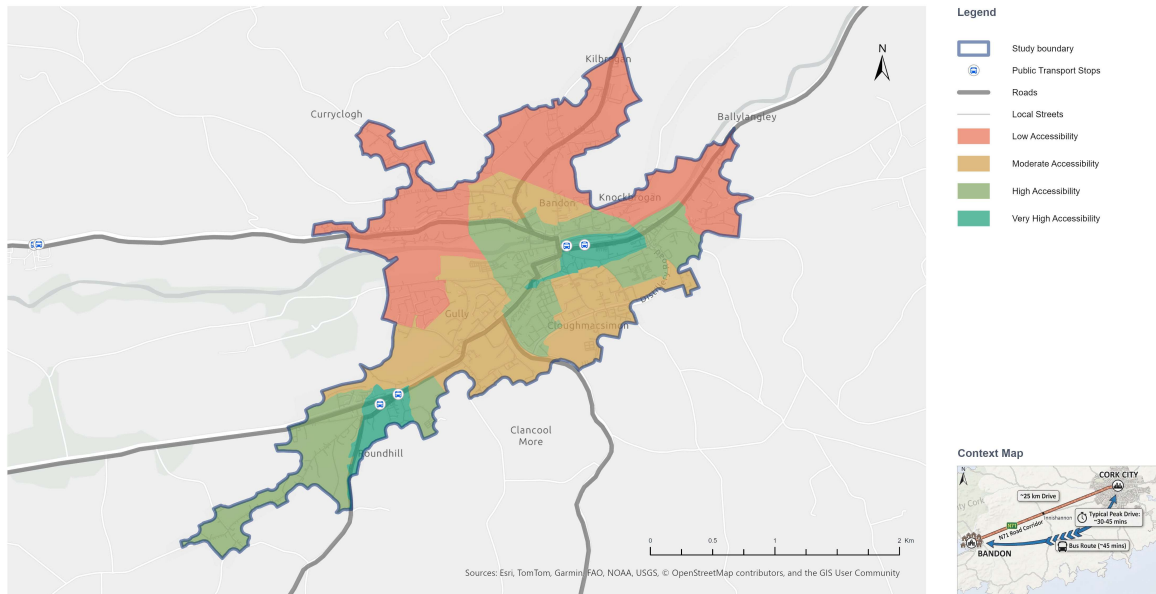
Key Message

Public transport accessibility extends outward from existing stops through the pedestrian network, with the strongest potential access concentrated around the established stop network and progressively weaker coverage towards areas beyond the longer walking thresholds.

4.3 Map 01.02 - Walking Distance Score

SEL - 002 01.02 | BANDON PUBLIC TRANSPORT ACCESSIBILITY - WALKING DISTANCE SCORE

Spatial accessibility to public transport stops based on walking distance



STUDY AREA

The walking-distance analysis evaluates accessibility to public transport stops across Bandon. Areas are classified into four accessibility levels based on walking distance, revealing how access varies from well-served areas near transport stops to locations farther from the network.

KEY MESSAGE

Walking distance creates a clear accessibility gradient across Bandon. Very high and high accessibility are concentrated around public transport stops, while moderate and low accessibility extend toward the town's outer areas. This highlights opportunities to improve public transport access in areas with longer walking distances.

DATA SOURCES

CSO - OpenStreetMap - NTA - Taillie Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map converts network-based walking distance to public transport into a relative accessibility score, classified from Low to Very High. It provides a continuous spatial interpretation of pedestrian access and highlights the transition between areas located close to public transport stops and areas requiring longer walking journeys.

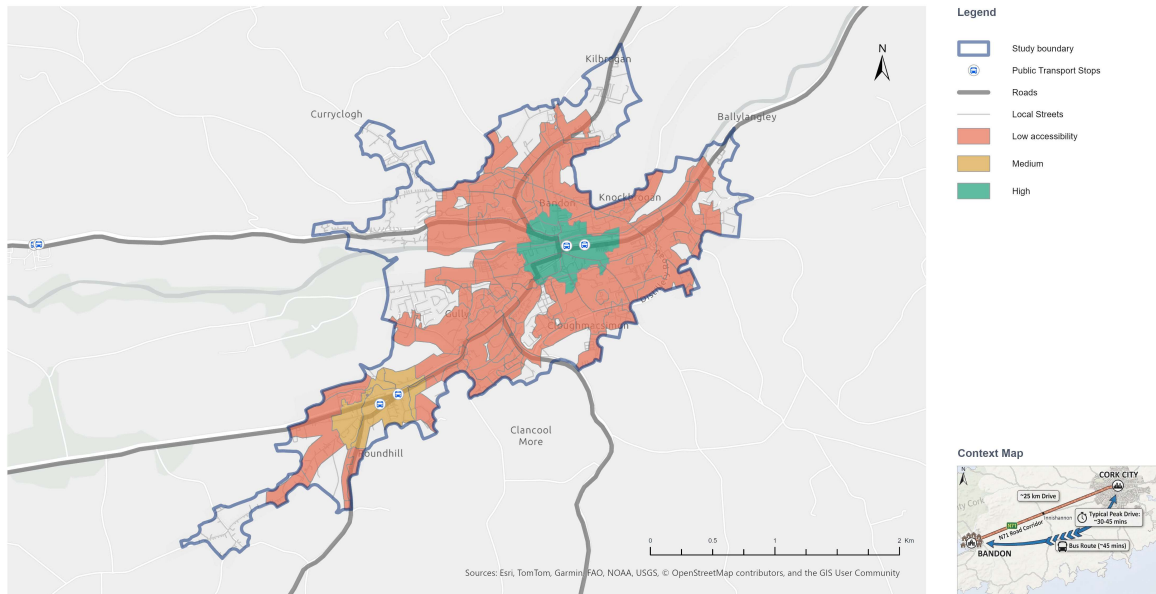
Key Message

Walking access follows a clear core-to-periphery gradient: areas closer to the public transport network have stronger accessibility, while longer walking distances contribute to weaker accessibility across peripheral parts of Bandon.

4.4 Map 01.03 - Peak-Time Service

SEL - 002 01.03 | BANDON PUBLIC TRANSPORT ACCESSIBILITY - PEAK-TIME SERVICE

Frequency-adjusted public transport accessibility during peak travel periods



STUDY AREA

The peak-time analysis assesses public transport accessibility across Bandon by combining proximity to stops with service frequency during periods of highest demand. The resulting classes show where residents have access to both nearby stops and more frequent services.

KEY MESSAGE

Peak-time accessibility is strongest around the central transport corridor, where stop proximity and service frequency overlap. Accessibility generally declines toward the outer areas, while the southwest shows a localized pocket of medium accessibility. This demonstrates that being close to a stop does not necessarily mean having high-quality service.

DATA SOURCES

CSO - OpenStreetMap - NTA - Taillie Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map introduces the temporal dimension of accessibility by combining spatial access to public transport with scheduled peak-time service frequency derived from GTFS data. It distinguishes locations where residents may be physically close to a stop from locations where that stop also provides a stronger level of scheduled service during peak travel periods.

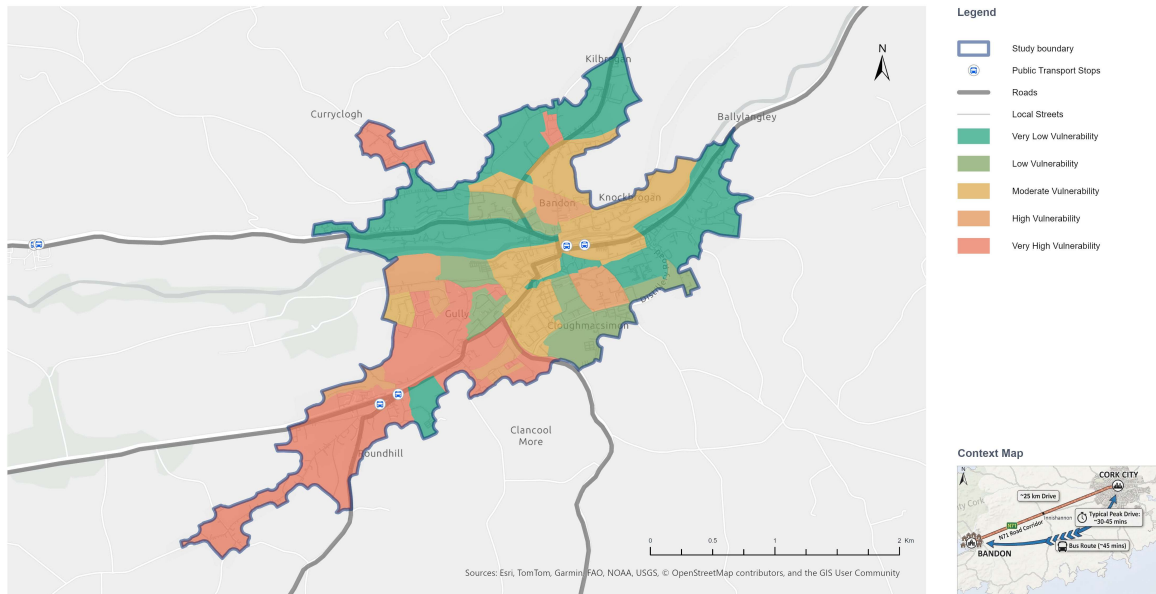
Key Message

Being close to a bus stop does not necessarily provide the same level of effective accessibility: the availability and frequency of scheduled peak-time services materially influence the usefulness of public transport.

4.5 Map 01.04 - Population Vulnerability

SEL - 002 01.04 | BANDON POPULATION VULNERABILITY

Census 2022 vulnerability patterns across the Bandon study area



STUDY AREA

The population vulnerability analysis assesses social and demographic vulnerability across the Bandon study area using Census 2022 data. The index combines multiple indicators to identify areas ranging from very low to very high vulnerability, highlighting spatial differences in population sensitivity and capacity to cope with change.

KEY MESSAGE

Population vulnerability is spatially uneven across Bandon, with higher vulnerability concentrated in several areas toward the south and southwest, while lower vulnerability is more prevalent around the central and northern areas. The pattern highlights neighbourhoods where targeted planning and social support may be most needed.

DATA SOURCES

CSO - OpenStreetMap - NTA - Taithe Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map represents the population-need dimension of the study using Census 2022 Small Area indicators relating to demographic, health and mobility, socio-economic and transport-dependence characteristics. The resulting vulnerability classification identifies areas with relatively lower or higher concentrations of population characteristics that may increase potential dependence on public transport or exposure to transport disadvantage.

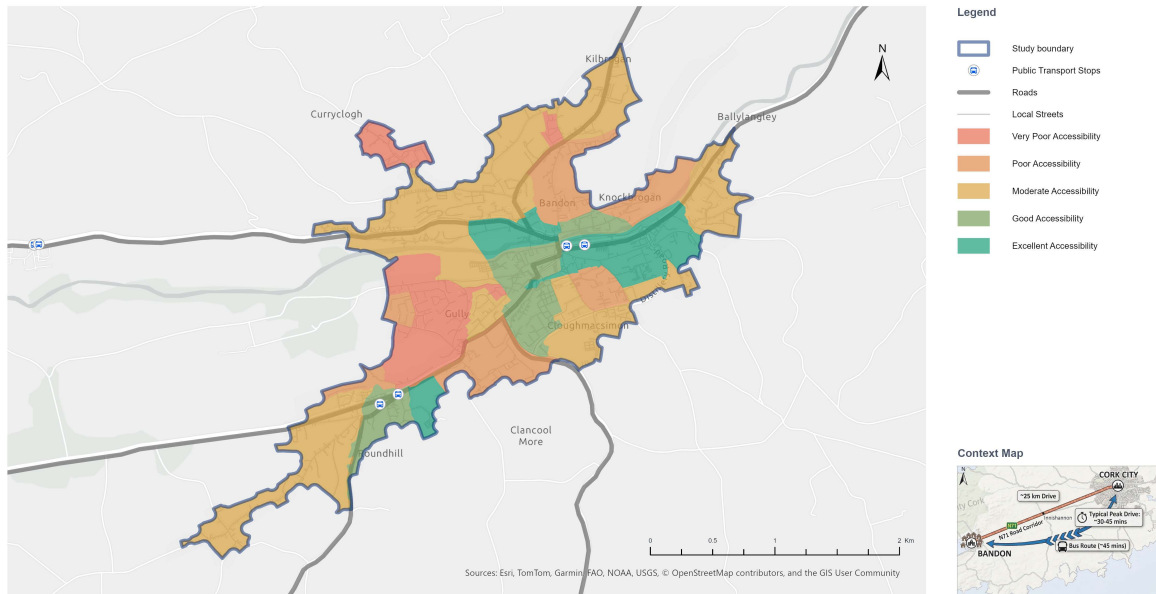
Key Message

Population vulnerability is spatially uneven across Bandon and does not necessarily coincide with the areas benefiting from the strongest public transport accessibility.

4.6 Map 01.05 - Composite Accessibility Index (CAI)

SEL - 002 01.05 | BANDON PUBLIC TRANSPORT COMPOSITE ACCESSIBILITY INDEX (CAI)

Integrated accessibility based on walking access and public transport service frequency



STUDY AREA

The CAI provides the clearest overall accessibility hierarchy. The map shows a relatively stronger accessibility structure around the central urban area, with poorer accessibility extending into peripheral areas. The five classes range from Very Poor to Excellent.

KEY MESSAGE

Accessibility is concentrated around Bandon's central area and public transport corridors. Improving pedestrian connections and public transport coverage in peripheral areas could help create a more balanced and connected accessibility network.

DATA SOURCES

CSO - OpenStreetMap - NTA - Taillie Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



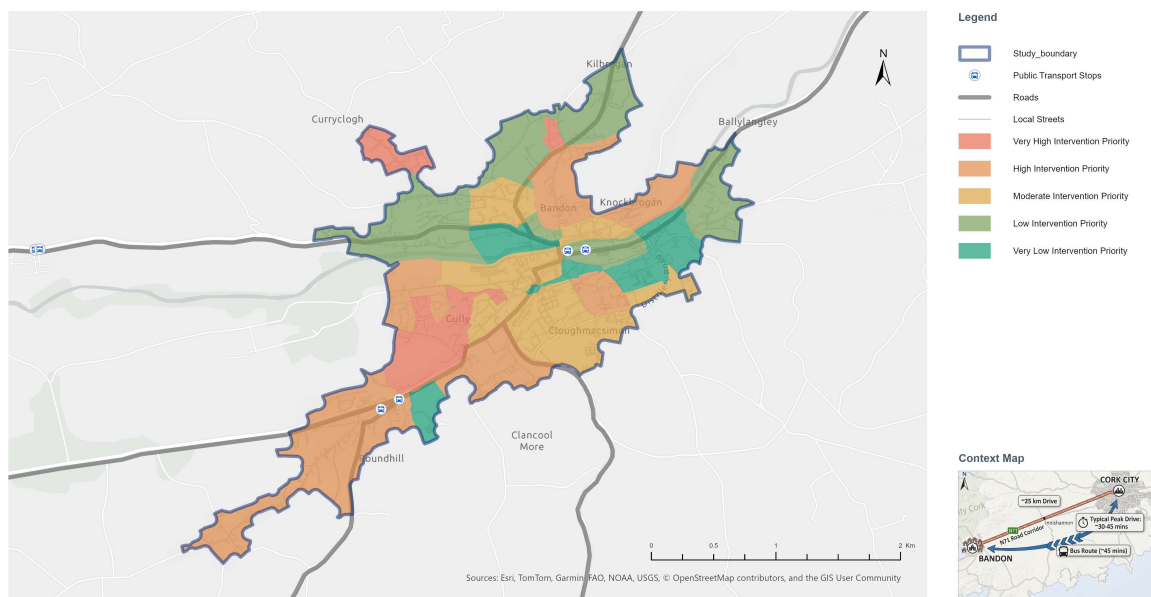
Description

The Composite Accessibility Index (CAI) integrates the two principal transport-access dimensions assessed in the study: network-based walking accessibility and scheduled service frequency. By combining physical access with service availability, the CAI provides an overall spatial hierarchy of public transport accessibility ranging from Very Poor to Excellent.

Key Message

The combined accessibility measure reinforces a pronounced core-to-periphery pattern, with stronger overall public transport accessibility around the central urban area and weaker conditions extending into several peripheral residential areas.

Identifying priority areas where public transport accessibility deficits overlap with population vulnerability



STUDY AREA

The assessment identifies areas across Bandon where public transport accessibility and population vulnerability overlap to create different levels of intervention priority. The analysis highlights locations where limited transport accessibility coincides with higher population vulnerability, while also distinguishing areas where lower vulnerability or stronger transport accessibility reduces the need for intervention.

KEY MESSAGE

Intervention priority is highest where very low public transport accessibility coincides with high population vulnerability. These areas should receive the greatest consideration for targeted transport and accessibility improvements, while areas with better accessibility relative to vulnerability represent progressively lower intervention priorities.

DATA SOURCES

CSQ · OpenStreetMap · NTA · Tailte Éireann

METHOD & GEOGRAPHY

Accessibility—Vulnerability Spatial Integration Model (AVSIM) · Irish Transverse Mercator (ITM)



This map identifies areas where public transport accessibility conditions intersect with population vulnerability to establish an equity-focused spatial priority framework. The model combines the CAI-derived accessibility deficit with the Census-based vulnerability measure, giving greater weight to population vulnerability. The resulting categories distinguish Critical and High Priority areas from locations where lower accessibility is associated with lower vulnerability or where accessibility is relatively strong.

Transport disadvantage is concentrated where weaker accessibility overlaps with higher population vulnerability; therefore, the highest-priority locations are not simply the least accessible areas, but those where accessibility deficits coincide with greater potential transport dependence.

Across the seven maps, the evidence develops from network coverage to spatial accessibility, service availability, population vulnerability and finally combined priority. The consistency of the mapped core-periphery pattern strengthens the interpretation that public transport accessibility in Bandon is spatially uneven.

5. Evaluation of Research Questions

The central analytical finding is that public transport accessibility cannot be adequately represented by a single distance measure. The Bandon map series demonstrates the importance of combining physical access, service availability and population characteristics.

- **Accessibility is More Than Distance:** Walking distance is an essential first component because residents must reach a stop before scheduled public transport can be used. However, the frequency-adjusted map demonstrates that accessibility also depends on service intensity. The CAI therefore provides a more complete representation of the transport system's spatial performance.
- **Walking Accessibility and Urban Form:** The walking catchments and distance score show the importance of the pedestrian network and settlement structure. The urban core is more closely connected to public transport stops, while peripheral residential areas are more likely to fall into weaker accessibility classes.
- **Temporal Accessibility:** Peak-time frequency introduces time into the accessibility framework. A transport stop should therefore be understood not simply as a point on a map, but as an access point with a schedule and service intensity.
- **Population Need vs. Service Provision:** The vulnerability layer provides a separate representation of population need. Comparing it with the CAI makes it possible to distinguish between areas that are poorly served but less vulnerable and areas where weaker accessibility coincides with greater vulnerability.
- **Composite Accessibility Index:** The CAI is the principal-integrated accessibility measure in the study. Its five-class hierarchy simplifies a multidimensional system into a spatially interpretable indicator while preserving the component maps for diagnostic purposes.
- **Accessibility-Vulnerability Hotspots:** The hotspot layer moves from measurement to prioritisation. Critical and High Priority areas are not simply low-accessibility areas; they represent the interaction between accessibility performance and population vulnerability.
- **Core-Periphery Accessibility Pattern:** The combined map evidence indicates a pronounced core-periphery structure. The central area is generally associated with stronger accessibility, while several peripheral areas exhibit lower accessibility and, in places, higher vulnerability.
- **Identifying Transport Disadvantage:** Transport disadvantage is best understood as a spatial relationship between the transport system and population characteristics. The hotspot map provides an operational method for identifying locations where this relationship warrants further investigation or intervention.

5.1 Research Question One - Viability for Cork City Commuters

Executive Finding: Public transport is a viable alternative for residents in the central corridor, but significantly less viable for peripheral neighbourhoods:

Spatial Zone	Viability Assessment	Key Factors
Central Corridor	High	Short walking distance, high peak service frequency, direct regional bus runs to Cork City
Intermediate Areas	Moderate	Reasonable walking access, but declining peak service availability
Peripheral Areas	Limited / Low	Long walking journeys, reduced service coverage, making private cars far more competitive

5.2 Research Question Two - Identifying Vulnerable Areas and Groups

Executive Finding: Several peripheral neighbourhoods combine low public transport accessibility with higher population vulnerability, marking them as key areas of transport disadvantage.

Affected Group	Primary Barrier	Planning Need
Older Adults (70+)	Reduced walking radius to bus stops	Highly accessible, localized stop placement
Disabled / Poor Health	Physical network barriers and topography	High-quality footpaths, crossings, and low-floor buses
Long-Term Unemployed	High transport costs or poor access to job centers	Affordable, frequent peak commuting services
Car-Less Households	Total reliance on active travel and public transit	High-density transit coverage in residential sectors

5.3 Spatial Typology of Bandon

- **System 1 (High Access / Lower Vulnerability):** Town core; well served by existing regional bus lines.
- **System 2 (Moderate Access / Mixed Vulnerability):** Transitional residential zones requiring selective service adjustments.
- **System 3 (Low Access / High Vulnerability):** Critical and High Priority Hotspots in peripheral estates where transport disadvantage is concentrated.

6. Policy and Planning Implications

6.1 Priority Intervention Areas

Critical Hotspot and High Priority areas should be considered candidate locations for further assessment. These locations warrant examination of walking infrastructure, stop placement, service frequency and the characteristics of the population concerned.

6.2 Pedestrian Infrastructure Improvements

Where poor accessibility is driven primarily by walking access, potential interventions include improved footpaths, crossings, pedestrian connections and safer links to existing public transport stops. The walking-catchment map provides a useful starting point for this assessment.

6.3 Service Frequency Optimisation

Where walking access is reasonable but peak-time service is weaker, service-frequency interventions may provide a more appropriate response than new infrastructure alone.

6.4 Bus Stop Expansion and Network Coverage

Peripheral areas outside the strongest walking-access zones may warrant assessment of additional stops, route adjustments or improved pedestrian connections. Such decisions require feasibility, demand and operational analysis beyond the current map series.

6.5 Accessibility Monitoring

The GIS workflow can be converted into a repeatable monitoring framework. Re-running the analysis as GTFS schedules, population data and pedestrian networks change would allow accessibility trends and emerging priority areas to be tracked.

6.6 Recommendations for Cork County Council and NTA

- Use the hotspot map as an initial screening layer for transport-investment and accessibility studies.
- Examine Critical Hotspot and High Priority areas using site-level pedestrian and service assessments.
- Review whether lower-accessibility peripheral areas require improved pedestrian connections, service changes or additional stops.
- Integrate scheduled-service accessibility analysis with wider active-travel and transport-demand evidence.
- Establish a repeatable GIS-based accessibility monitoring process using versioned GTFS and population datasets.

7. Limitations and Future Research

The map outputs are analytical indicators rather than direct observations of individual travel behaviour. Their interpretation depends on the input data, network model, classification rules and assumptions used in the processing workflow.

7.1 Limitations

- **Walking Assumptions:** Network accessibility depends on the completeness and connectivity of the pedestrian network and on the assumed walking speed. The supplied outputs document the walking-time thresholds but not the precise speed parameter.
- **GTFS Limitations:** GTFS represents scheduled service. It does not necessarily capture delays, cancellations, temporary diversions, crowding or real-time operational conditions.
- **Census Temporal Limitations:** Population vulnerability is based on Census 2022 and therefore represents the population reference available to the analysis rather than a continuously updated population estimate.
- **Accessibility Does Not Measure Demand:** The accessibility index measures spatial conditions of access to public transport. It does not measure actual rider ship, trip demand, individual travel behaviour or unmet demand.
- **Interpretation of Composite Indices:** Composite indices simplify several dimensions into a single score or class. They are useful for comparison and prioritisation, but should be interpreted alongside the underlying component maps.
- **Appropriate Use of Hotspot Results:** Hotspots should be treated as screening and prioritisation evidence. A Critical Hotspot does not, by itself, establish that a specific intervention is required or that a location has objectively unmet demand. Local validation, stakeholder knowledge and operational feasibility assessment remain necessary.

7.2 Future Research Priorities

- **GTFS - Realtime Accessibility:** Integrate real-time or operational data to move beyond scheduled accessibility and capture service reliability and disruption.
- **Accessibility by Time of Day:** Extend the frequency analysis beyond peak periods to compare morning, inter-peak, evening and weekend accessibility.
- **Accessibility for Schools and Healthcare:** Measure accessibility to specific essential destinations and examine whether vulnerable populations have adequate access to them.
- **Accessibility for Active Travel:** Integrate walking and cycling infrastructure quality, safety and connectivity into the accessibility framework.
- **Integration into Ireland Quality of Life Index:** Develop transport accessibility as a spatial quality-of-life dimension that can be integrated with the wider Ireland Small Area Quality of Life Index framework.
- **Replication Across Irish Towns:** The methodology could be tested in other towns to assess how accessibility patterns vary by settlement structure, service provision and population vulnerability, such as Clonakilty, Skibbereen, Bantry, Midleton, and Mallow.

8. Conclusion

The Bandon Public Transport Accessibility Analysis demonstrates how GIS can integrate transport network accessibility, scheduled service frequency and population vulnerability into a coherent small-area evidence framework.

8.1 Summary of Findings

- Public transport accessibility is spatially uneven across Bandon.
- Walking accessibility is strongest around areas close to public transport stops and weakens toward peripheral areas.
- Peak-time service frequency introduces an important temporal dimension beyond physical distance.
- Population vulnerability is spatially differentiated and does not necessarily coincide with the best-served areas.
- The Composite Accessibility Index reveals a clear accessibility hierarchy.
- The Accessibility-Vulnerability Hotspot analysis identifies locations where lower accessibility and higher vulnerability combine to indicate greater potential priority.

8.2 Research Contribution

The principal contribution is methodological: the study demonstrates a reproducible spatial workflow for assessing public transport accessibility in a smaller Irish town using openly available spatial data, GTFS schedules and Census 2022 Small Area information. It also provides a framework for linking transport accessibility to wider social-equity and quality-of-life analysis.

8.3 Final Reflection

Accessibility should be understood as an essential component of inclusive mobility and quality of life. In Bandon, the spatial evidence shows that transport accessibility varies across the settlement and that the most meaningful planning questions emerge where accessibility conditions intersect with population vulnerability.

References

Category	Examples / Source Framework
Accessibility Theory	Hansen (1959); Geurs & van Wee (2004)
Transport Equity	Lucas (2012); Martens (2016)
GIS Accessibility	ESRI Network Analyst literature
Irish Transport Policy	NTA; Department of Transport
Census & Spatial Analysis	CSO publications

Appendix A. Data Sources and Reproducibility Register

This appendix records the principal datasets represented in the current study. Exact dataset versions and extraction dates should be added when the production metadata is finalised.

Dataset	Version / Date	Purpose
CSO Census 2022 Small Areas	2022	Population and socio-demographic indicators.
Bus Éireann / NTA GTFS	GTFS feed 2026	Stops, routes, and stop times.
OpenStreetMap	Extract 2026	Walking street network and local spatial context.
Administrative / Small Area Boundaries	Tailte Éireann / CSO	Study and reporting geography.
Spatial Evidence Lab PostGIS Database	Version 1.0	Processed accessibility and vulnerability layers.

Appendix B. Map Production Register

The following register is the authoritative map naming structure for the SEL-002 deliverables.

Map ID	Map Title	Subtitle	File Name
01.00	Bandon Public Transport Accessibility Study Area	Study boundary, public transport stops and local road network	SEL-002_01.00
01.01	Bandon Public Transport Accessibility - Walking Catchments	Areas within 5, 10 and 15 minutes walking distance of public transport stops	SEL-002_01.01
01.02	Bandon Public Transport Accessibility - Walking Distance Score	Spatial accessibility to public transport stops based on walking distance	SEL-002_01.02
01.03	Bandon Public Transport Accessibility - Peak-Time Service	Frequency-adjusted accessibility to public transport during peak travel periods	SEL-002_01.03
01.04	Bandon Population Vulnerability	Spatial distribution of population vulnerability based on Census 2022	SEL-002_01.04
01.05	Bandon Composite Public Transport Accessibility Index	Integrated assessment of walking access and public transport service frequency	SEL-002_01.05
01.06	Bandon Public Transport Accessibility - Equity - focused vulnerability hotspots	Areas where public transport accessibility and population vulnerability indicate differing levels of spatial priority	SEL-002_01.06

Document Control

Project: SEL-002 - Bandon Public Transport Accessibility Analysis